Many daughters PAP

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In this pre-analysis plan, we describe how we plan to use SOEP data to test the following four hypothesis:

- **Hypothesis 1 (H1):** Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality,
  (X for respondents with daughters) > (X for respondents without daughters).
- **Hypothesis 2 (H2).** Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

- **Hypothesis 3 (H3)**. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).

- **Hypothesis 4 (H4)**. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality,(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters).

Throughout, we

- set all special answer codes (e.g. -1 = missing, -2 = does not apply) to missing
- combine variables in from different survey waves even if they have slightly different wording
- include data from all years if an outcome variable if available in multiple years. For example, if an outcome is available in two years, each of those is treated as separate observation. In those cases, we also include survey-year (waive) dummies.

Testing H1

To test H1, we plan to estimate the following model:

$$GEattitude_{ift} = \alpha_0 + \alpha_1 daughters_{ift} + X_{ift}\gamma + u_{ift},$$

where $GEattitude_{ift}$ for respondent i of household f in survey year t is a variable based on these two survey items:

1. *Women have the same rights as men*, where possible answers range from 0 (Should definitely not happen in a democracy) to 10 (Should definitely happen in a democracy).
2. *At home, the husband should have the final say*, where possible answers range from 0 (Do not agree at all) to 7 (I totally agree).

To create the $GEattitude_{ift}$ variable, we

- Reverse item 2 (so that higher values indicate less agreement with the statement)
- Rescale reversed item 2 to range from 0 to 10 (rescaled_2 = (reversed_2/7)*10)
- Take the average of original item 1 and the reverse and rescaled item 2.
- For respondents with data on only one of the two items, we use the available item to avoid missing values.

Our independent variable of interest is $daughters_{ift}$, which is a dummy variable indicting whether the respondent i of household f has at least one daughter (=1) or only sons (=0) in survey year t. This variable is missing for respondents who do not have any children.

Control variables X_{ift}:

- male dummy (1 if male, 0 if female),
- German dummy (1 if German born, 0 if not).
- age of respondent in years
- age of the first child
- last available level of education before the birth of the first child (using dummy variables, one for each available category minus a base category)
- survey year dummies

Sample restrictions: We limit our sample to respondents with children. This happens mechanically, because $daughters_{ift}$ is missing for respondents without children. We do no plan to do any other sample restrictions.

Our parameter of interest is α_1, which shows the causal effect of having at least one daughter (compared to only having sons) on gender equality attitudes (measured as described above).

We estimate α_1 with ordinary least squares regression. We cluster our standard error at the household level (one cluster per household).

State code:

$$reg\ GEattitude\ daughters\ male\ german\ AgeRespondent\ AgeChild$$
$$i.education\ i.year, cluster(household\ ID)$$

We would confirm H1 if the estimate of α_1 is positive and the corresponding p-value for a two-sided test is smaller than 0.05.

Testing H2

To test H2, we plan to estimate the following model:

$$Leftleaning_{ift} = \beta_0 + \beta_1 daughters_{ift} + X_{ift}\delta + \epsilon_{ift},$$

where $Leftleaning_{ift}$ for individual i of household f in survey year t is based on this survey item:

- *In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?,* where possible answers range from 0 (very left) to 10 (very right).

We reverse this variable so that higher values indicate more left-leaning political views. Daughters is defined as above and X_{ift} includes the same control variables as above.

The parameter of interest is β_1 which represents the causal effect of having at least one daughter (compared to only having sons) on how left-leaning respondents' political views are.

We estimate β_1 with an instrumental variable approach, using an indicator whether the first-born child is female or male ($firstfemale_{if}$) as instrument for $daughters_{ift}$. We use this approach because we are concerned about daughters being endogenous. For example, people who have more left-leaning political views might also have a stronger preference for daughters and therefore have more children until they have at least one daughter (those concerns are valid for H1 as well, but the expected sample size is too small to make IV a suitable option). We use the gender of the first-born child is instrument, because it is plausibly unrelated to other factors influencing political preferences.

We use the Stata command "tf" by Lee et al. (2022) to estimate β_1 and conduct inference.

Stata code: $tf\ Leftleaning\ (daughters = first_female)\ male\ german\ AgeRespondent\ AgeChild$

$$i.education\ i.year, cluster(household\ ID)$$

We would confirm H2 if the estimate of β_1 is positive and the "tF adjusted 95-Confidence interval" does not include zero.

In the case of the confidence interval is not bounded, we will revert to estimating the model by OLS.

Stata code: $reg\ Leftleaning\ daughters\ male\ german\ AgeRespondent\ AgeChild$

$$i.education\ i.year, cluster(household\ ID)$$

We would confirm H2 if the estimate of β_1 is positive and the corresponding p-value for a two-sided test is smaller than 0.05.

Testing H3

To test H3, we plan to estimate the following model:

$$GEoutcome_{ift} = \gamma_0 + \gamma_1 daughters_{ift} + X_{ift}\phi + \eta_{ift},$$

where $GEoutcome_{ift}$ for individual i of household f in survey year t is based on this survey question: *What does your everyday life look like in the current situation? How many hours per normal work day do you spend on the following activities?* We define $GEoutcome_{if}$ as the sum of the reported number of hours for the following two categories:

1. physical activities (sports, fitness, gymnastics)
2. other recreational activities or hobbies

Daughters is defined as above and X_{ift} includes the same control variables as above.

The parameter of interest is γ_1 which represents the causal effect of having at least one daughter (compared to having only sons) on respondents' Gender equality outcome (as measured with time spend on physical activities and other hobbies).

We estimate γ_1 with ordinary least squares regression. We cluster our standard error at the household level (one cluster per household).

Stata code: $reg\ GEoutocme\ daughters\ male\ german\ AgeRespondent\ AgeChild$

$$i.education\ i.year, cluster(household\ ID)$$

We would confirm H3 if the estimate of γ_1 is positive and the corresponding p-value for a two-sided test is smaller than 0.05.

Testing H4

To test H4, we plan to estimate the following model:

$$GEattitude_{ift} = \pi_0 + \pi_1 daughters_{ift} + \pi_2 male_{if} + \pi_3 daughters_{ift} \times male_{if} + M_{ift}\varphi + \upsilon_{ift},$$

where $GEattitude_{ift}$ and $daughters_{ift}$ are the same variables from H1 for individual i of household f in survey year t, and $male_{if}$ is a dummy variable indicating respondent is a male (1 if male, 0 if female),

Our independent variable of interest is $daughters_{ift} \times male_{if}$, which is an interaction term between dummy variable for having at least one daughter and dummy variable for the respondent being male.

Control variables M_{ift}:

- German dummy (1 if German born, 0 if not).
- age of respondent in years
- age of respondent when having the first child
- last available level of education before the birth of the first child (using dummy variables, one for each available category minus base category)
- survey year dummies

Sample restrictions: We limit our sample to respondents with children.

Our parameter of interest is π_3, which shows the difference in the effect of having at least one daughter (compared to only having sons) on gender equality attitudes for male respondents compared to the same effect for female respondents. It is a difference-in-difference effect.

We estimate π_3 with ordinary least squares regression. We cluster our standard error at the household level (one cluster per household).

State code: xi: $reg\ GEattitude\ i.daughters * i.male$
$german\ AgeRespondent\ AgeChild\ i.education\ i.year, cluster\ (household\ ID)$

We would confirm H1 if the estimate of π_3 is positive and the corresponding p-value for a two-sided test is smaller than 0.05.

References

Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t-Ratio Inference for IV." American Economic Review, 112 (10): 3260-90.